Table Creation

```
-- Department Table
CREATE TABLE Department (
   dept_id INT PRIMARY KEY,
   dept_name VARCHAR(50)
);

-- Employee Table
CREATE TABLE Employee (
   emp_id INT PRIMARY KEY,
   emp_name VARCHAR(50),
   dept_id INT,
   salary DECIMAL(10,2),
   hire_date DATE,
   FOREIGN KEY (dept_id) REFERENCES Department(dept_id)
);

-- Project Table
CREATE TABLE Project (
   proj_id INT PRIMARY KEY,
   proj_name VARCHAR(50),
   dept_id INT,
   FOREIGN KEY (dept_id) REFERENCES Department(dept_id)
);

-- Employee_Project Mapping Table
CREATE TABLE Emp_Project (
   emp_id INT,
   proj_id INT,
   PRIMARY KEY(emp_id, proj_id),
   FOREIGN KEY (emp_id) REFERENCES Employee(emp_id),
   FOREIGN KEY (proj_id) REFERENCES Project(proj_id)
);
```

2. Insert Data

```
-- Department Data
INSERT INTO Department VALUES
(1, 'HR'),
(2, 'IT'),
(3, 'Finance'),
(4, 'Marketing');
```

```
-- Employee Data
INSERT INTO Employee VALUES
(101, 'Alice', 1, 50000, '2020-01-15'),
(102, 'Bob', 2, 70000, '2019-03-10'),
(103, 'Charlie', 2, 80000, '2018-06-20'),
(104, 'David', 3, 60000, '2021-09-01'),
(105, 'Eva', 4, 55000, '2020-11-12');

-- Project Data
INSERT INTO Project VALUES
(201, 'Payroll System', 3),
(202, 'Website Revamp', 2),
(203, 'Recruitment Portal', 1),
(204, 'Ad Campaign', 4);

-- Employee_Project Mapping
INSERT INTO Emp_Project VALUES
(101, 203),
(102, 202),
(103, 202),
(103, 201),
(104, 201),
(105, 204);
```

1. **List employees with their department names**
2. List employees working on projects with project names
3. Find employees in IT department working on projects
4. List all projects and their managing departments
5. Employees not assigned to any project
6. **List all employee names and department names**
7. Employees and Departments with common names
8. Employees not in IT department
9. Projects handled by Finance but not by IT
10. List of all departments + projects together
11. Find total employees in each department
12. Find average salary of employees
13. Find maximum salary in IT department
14. Find minimum salary in Finance department
15. Find total salary expense per department
16. Find number of employees working on each project
17. Find departments having more than 1 employee
18. Find average salary per department
19. Find employees hired after 2020
20. Find highest paid employee per department